

Conformer-based On-device Streaming Speech Recognition with KD Compression and Two-pass Architecture

Jinhwan Park^{1*}, Sichen Jin^{1*}, Junmo Park^{1*}, Sungsoo Kim^{1*}, Dhairya Sandhyana¹, Changheon Lee², Myoungji Han², Jungin Lee², Seokyeong Jung², Changwoo Han¹, Chanwoo Kim¹

¹*Samsung Research, South Korea*, ²*AI R&D Group, Samsung Electronics, South Korea*

SLT-2022, 1-2-8-ASR

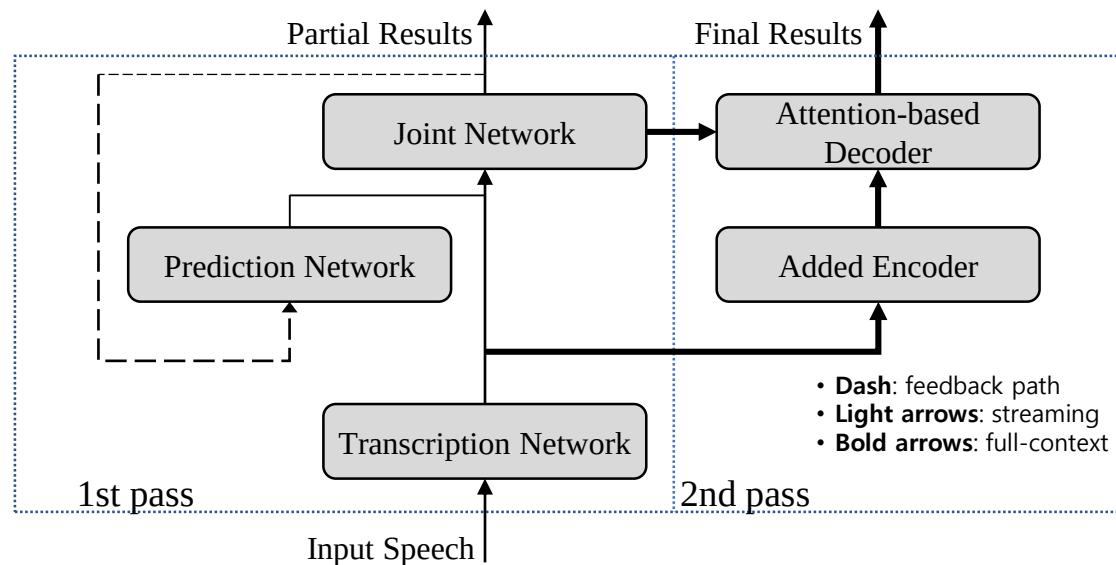
Mon 9 Jan, 2023 - Afternoon session (13:30 -15:00)



► Model Architecture

Two-pass E2E ASR model

Architecture of two-pass E2E ASR model



- 1st pass: Conformer-transducer
- 2nd pass: attention-based encoder decoder (rescoring / beam search)
- In-house training toolkit and training data

Model compression

- We use Knowledge Distillation (KD) for the compression of the 2nd pass model
 - Set different compression rates for the recurrent and the non-recurrent parameters of the 2nd pass model.
- The E2E model is trained in 3 stages where the losses are defined as follows:

1st pass streaming Conformer-T model: L_T

2nd pass teacher model : $L_T + \lambda_1 L_{2nd}$

2nd pass student model + KD : $L_{2nd} + \lambda_2 L_{dist}$

► Decoder Implementation

Language model shallow fusion

- **General-domain Language Model (LM)**

- Transformer-XL based language model
- Integrated with the ASR result by shallow fusion

$$\begin{aligned} \log P(y_u | x_{1:t}, y_{1:u-1}) \\ = \log P_{\text{ASR}}(y_u | x_{1:t}, y_{1:u-1}) + \sum_i \lambda_i \log P_{\text{LM}_i}(y_u | y_{1:u-1}) \end{aligned}$$

- **Domain-specific LM biasing**

- A domain-specific LM is built on-device from the template and NEs of the target domains.
- The probabilities computed from the LMs and the E2E models are interpolated at each beam search step using shallow fusion.

Decoding techniques

- **Multi-stage spell correction**

- Spell correction changes wrongly recognized NEs based on a multi-level string matching technique.
- Combines string matching (edit distance) scores at word, grapheme and phoneme levels

- **Decoding of unintended silence**

- Sometimes unwanted words are misrecognized
- Length normalization → accuracy degraded
- Solution: pruning low confident beams

- **Decoding of long-form speech**

- Time-based segmentation approach → deletion error is relatively high
- Solution: streaming energy-based segmentation

* NE: Named Entity

► Experimental Results

Experimental results

■ Model compression with KD

Model	# Params.	WER	• WER: Word Error Rate (%) • γ_r: compression ratios for recurrent • γ_f: compression ratios for non-recurrent • (# Params) for the 2nd pass model
(a) Conformer-T	37M	10.19	
(b) (a) + rescoring	102M (65M)	9.78	
(c) (b) + KD			
(c-1) $\gamma_r = 0.6, \gamma_f = 0.6$	66M (29M)	9.85	
(c-2) $\gamma_r = 0.7, \gamma_f = 0.7$	60M (23M)	9.96	
(c-3) $\gamma_r = 0.8, \gamma_f = 0.6$	57M (20M)	9.80	
(c-4) $\gamma_r = 0.9, \gamma_f = 0.7$	40M (13M)	10.19	
(d) (c-3) + more train steps	57M (20M)	5.65	
(e) (d) w/o TFXL LM	57M (20M)	6.54	

■ WER (%) on the domain-specific test set

Model	Contact	General	※ Biasing: biased with n-gram LM ※ Spell: applied spell corrector
Conformer-T + rescoring	11.36	5.65	
+ <i>Biasing</i>	9.89	5.96	
+ <i>Biasing</i> + <i>Spell</i> (syllable)	7.23	5.99	
+ <i>Biasing</i> + <i>Spell</i> (multi)	4.42	6.03	

Experimental results

■ Comparison of MoChA, RNN-T, and Conformer-T

Model	# Params.	WER
MoChA*	40M	9.33
RNN-T	73M (-)	8.72
RNN-T + rescoring	140M (67M)	8.24
Conformer-T	37M (-)	5.92
Conformer-T + rescoring	57M (20M)	5.65

■ Real-time factor

Model	xRT
RNN-T + rescoring	0.375
Conformer-T + rescoring	0.143

※ Measured on Galaxy Note 10+ device. Only a single core of CPU is used for measurement.

* MoChA: Monotonic Chunkwise Attention

**Please visit our poster in 1:30PM, 9 Jan
for more details!**

SLT-2022, 1-2-8-ASR

Mon 9 Jan, 2023 - Afternoon session (13:30 -15:00)

